Khushal Nirwal

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EDUCATION

Kalinga Institute of Industrial Technology *July 2023 – Present*
B.Tech in Computer Science Engineering | **CGPA: 8.92**

Delhi Public School, Rohini *2021 – 2023*
Senior Secondary Education | **AISSCE: 86%**

EXPERIENCE

AI Engineer Intern | *Deloitte South Asia, Gurugram, Haryana* *July 2026 – August 2026*

- Shipped AI features for an internal tool Deloitte Tax uses to file TDS/TCS returns.
- Built a **RAG pipeline on AWS Bedrock** to find each transaction's TDS section, using **Amazon Titan Embeddings** to search tax provisions and **Amazon Nova Lite** to pick the TDS section code.
- Paired the RAG pipeline with **rule-based checks** on G/L account names, raising accuracy from **66% to 94%** on ∼500 labeled transactions.
- Designed the end-to-end G/L preprocessing pipeline with per-company normalization layers, turning unstructured ledger data into a validated, deduplicated schema.

Software Engineer Intern | *Teemo.ai, NYC (Remote)* *February 2026 – June 2026*

- Built backend services for an AI-powered education platform, **solely owning** the At-Risk Student Detection system, which combines 5 behavior signals into one score and sorts students into 3 risk levels, with **∼100% test coverage**.
- Co-developed an **AI photo-grading pipeline** scoring handwritten worksheets via LLM, and built the AI Tutor backend behind a **pluggable provider interface**, decoupling tutoring logic from any single LLM vendor.

PROJECTS

Sentinel AI - 1D CNN-Based Network Intrusion Detection System [Link]

- Trained a **1D CNN** classifying network flows into 11 classes (benign + 10 attack families: DDoS, PortScan, Botnet, etc.) on CICIDS2017 across 78 features, reaching **99.48% accuracy and 97.67% macro-F1** on 47,912 test flows.
- Cut false alarms via class balancing and dropout, clearing 98.97% of benign traffic (**∼1% false-positive rate**) at 98.29% average precision and 97.20% average recall across all 11 classes.
- Shipped a containerized **FastAPI + React** app with single and batch endpoints (up to 5,000 flows/request), **SHAP** per-feature explanations and a live demo WebSocket alert stream, at **∼1,800 flows/sec** and ∼155 ms p95 latency on CPU.
- **Tools:** Python, TensorFlow/Keras, scikit-learn, SHAP, NumPy, pandas, FastAPI, React, WebSockets, Docker

Swiggy Jugaad - Agentic AI Grocery Assistant on Swiggy MCP [Link]

- Built an AI agent on Swiggy's **Instamart MCP server** that infers what is left in a household's kitchen from order history, ranks recipes by ingredients likely on hand, and places a **real Instamart order** for only the missing items.
- Modelled pantry stock as a **Weibull (k=2) decay curve** per item, scaled sublinearly by household size, with a per-item pace multiplier learned from user corrections so the agent asks less over time.
- Wrote a **framework-free tool-calling agent loop** on NVIDIA NIM with model fallback, streamed to a web UI over **Server-Sent Events**, with checkout guardrails enforced in code (split stage/confirm tools, session-bound user ID, Rs. 1000 cap).
- Mapped raw SKU names to a **67-ingredient vocabulary** via batched LLM calls and a global SKU cache; **78 offline tests**.
- **Tools:** Python, FastAPI, MCP, OAuth 2.1 + PKCE, NVIDIA NIM, OpenAI SDK, PostgreSQL, Pydantic, SSE, pytest

TECHNICAL SKILLS

GenAI & LLMs: RAG, AWS Bedrock, LangChain, AI Agents, MCP, LoRA Fine-Tuning, HF Transformers, LangGraph
ML & Deep Learning: PyTorch, TensorFlow/Keras, scikit-learn, Classification, Clustering, Regression, SHAP
Data: NumPy, Pandas, EDA, Data Preprocessing, Feature Engineering, Matplotlib, Seaborn
Programming: Python, Java, SQL, C
Backend & Databases: FastAPI, REST APIs, SSE, PostgreSQL, SQLite, Unit Testing (pytest)
Infrastructure & Tools: Linux (daily driver, 2.5+ years), Docker, AWS, Git, Hugging Face Spaces, Koyeb
Coursework: Data Structures & Algorithms, OOP, DBMS, OS, Computer Networks, Probability & Statistics, Linear Algebra

CERTIFICATIONS & ACHIEVEMENTS

- **AWS Certified AI Practitioner** (Foundational) — Amazon Web Services
- **Stanford Statistical Learning Certification** — Stanford Online
- **Top 60 of 4000+ teams** — Deloitte Hacksplosion 2026 (national, top 1.5%)
- **400+ problems solved** on LeetCode and Codeforces